

Total No. of Questions : 8]

SEAT No. :

P806

[5870]-1126

[Total No. of Pages : 2

**T.E. (Computer Engineering)**  
**THEORY OF COMPUTATIONS**  
**(2019 Pattern) (Semester-I) (310242)**

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.
- 2) Figures to the right side indicate full marks.
- 3) Neat diagrams must be drawn wherever necessary.
- 4) Assume suitable data, if necessary.

- Q1) a)** Write a grammar G for generating the language [9]  
i)  $L = \{w \text{ belongs to } \{a,b\}^* \mid w \text{ is an even length palindrome with } |w| > 0\}$   
ii) Set of odd length strings in  $\{0,1\}^*$  with middle symbol '1'

- b)** Simplify the following grammar [9]  
 $S \rightarrow 0A0 \mid 1B1 \mid BB$   
 $A \rightarrow C$   
 $B \rightarrow S \mid A$   
 $C \rightarrow S \mid \epsilon$

OR

- Q2) a)** Reduce the following grammar to Greibach Normal form. [9]  
 $S \rightarrow AA \mid 0$   
 $A \rightarrow SS \mid 1$   
**b)** Construct a DFA for the following left linear grammar. [9]  
 $S \rightarrow B1/A0/C0$   
 $B \rightarrow B1/1$   
 $A \rightarrow A1/B1/C0$   
 $C \rightarrow A0$

- Q3) a)** Construct a context free grammar which accepts  $N(A)$ , where [9]  
 $A = (\{q_0, q_1\}, \{0,1\}, \{Z_0, Z\}, \delta, q_0, Z_0, \phi)$  where  $\delta$  is given by  
 $\delta(q_0, 1, Z_0) = \{(q_0, ZZ_0)\}$   
 $\delta(q_0, \epsilon, Z_0) = \{(q_0, \epsilon)\}$   
 $\delta(q_0, 1, Z) = \{(q_0, Z Z)\}$   
 $\delta(q_0, 0, Z) = \{(q_1, Z)\}$   
 $\delta(q_1, 1, Z) = \{(q_1, \epsilon)\}$   
 $\delta(q_1, 0, Z_0) = \{(q_0, Z_0)\}$

P.T.O.

- b) Construct a PDA that accept the language generated by grammar [8]  
 i)  $S \rightarrow 0S1 \mid A, A \rightarrow 1A0 \mid S \mid \epsilon$   
 ii)  $S \rightarrow aABB \mid aAA, A \rightarrow aBB \mid a, B \rightarrow bAA \mid A$

OR

- Q4)** a) What is NPDA? Construct a NPDA for the set of all strings over  $\{a,b\}$  with odd length palindrome: [9]  
 b) Design a push down automaton to recognize the language generated by the following grammar: [8]

$$S \rightarrow S + S \mid S \square S \mid 4 \mid 2$$

Show the acceptance of the input string  $2 + 2*4$  by this PDA.

- Q5)** a) What is a Turing Machine? Give the formal definition of TM. [9]  
 Design a TM that replaces every occurrence of abb by baa.  
 b) What are the different ways for extension of TM? Explain. [9]  
 Design TM for language  $L = \{a^i b^j \mid i < j\}$

OR

- Q6)** a) What is TM? Design TM to check well formedness of Parenthesis. Expand the transition for  $(())()$  [9]  
 b) Elaborate the following terms [9]  
 i) Universal Turing Machine (UTM)  
 ii) Recursively Enumerable Languages  
 iii) Halting Problem of Turing Machine

- Q7)** a) Justify "Halting Problem of Turing machine is undecidable". [9]  
 b) Define the Class P and Class NP and Problem with their example in detail. [8]

OR

- Q8)** a) Explain Satisfiability Problem and SAT Problem and comment on NP Completeness of the SAT Problem. [9]  
 b) What do you mean by polynomial time reduction? Explain with suitable example. [8]



Total No. of Questions : 8]

PA-1442

SEAT No. :

[Total No. of Pages : 3

[5926]-58

T.E. (Computer Engg.)

THEORY OF COMPUTATION

(2019 Pattern) (Semester-I) (310242)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3. or Q4, Q5 or Q6, and Q7 or Q8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figure to the right indicate full marks.
- 4) Assume suitable data if necessary.

**Q1) a)** Convert the following grammar to Chomsky Normal form (CNF) [9]

$S \rightarrow a \mid aA \mid B$

$A \rightarrow aBB \mid \epsilon$

$B \rightarrow Aa \mid b$

**b)** Convert the following grammar to GNF. [9]

$S \rightarrow XB \mid AA$

$A \rightarrow a \mid SA$

$B \rightarrow b$

$X \rightarrow a$

OR

**Q2) a)** Show that the following grammar is ambiguous. [6]

$S \rightarrow iCtS$

$S \rightarrow iCtSes$

$S \rightarrow a$

$C \rightarrow b$

**b)** Convert the following grammar to chomsky normal form (CNF) [6]

$G = (\{S\}, \{a, b\}, P, S)$

$P = \{S \rightarrow aSa \mid bSb \mid a \mid b \mid aa \mid bb\}$

P.T.O.

- c) Consider the following grammar. [6]

$E \rightarrow E + E \mid E - E \mid id$

Derive the string  $id-id*id$  using

- Leftmost derivation
- Rightmost derivation.

- Q3) a)** Find the transition rules of PDA for accepting a language

$L = \{w \in \{a,b\}^* \mid w \text{ is of the form } a^n b^n \text{ with } n \geq 1\}$  through both empty stack and final state and demonstrates the stack operation for the string  $aaabbb$ . [9]

- b) Design a PDA for accepting a language  $\{a^n b^{2n} \mid n \geq 1\}$  [9]

Simulate this PDA for the input string “ $aaabbbbbb$ ”.

OR

- Q4) a)** Design a PDA for accepting a language  $\{0^n 1^m 0^n \mid m, n \geq 1\}$ .

Simulate this PDA for the input string “ $00111100$ ”. [9]

- b) Construct a PDA for  $L = \{0^m 1^m 2^m 3^n \mid m, n \geq 0\}$  [6]

- c) Compare FA and PDA. [3]

- Q5) a)** Write a short note on Halting problem of Turing machine. [4]

- b) Design a Turing Machine for the following language by Considering transition table and diagram. [9]

i) TM That erases all non blank symbols on the tape where the sequence of non blank symbols does not contain any blank symbol B in between.

ii) TM that find 2's complement of a binary machine.

- c) Design a Turing Machine that reads a string representing a binary number and erases all leading 0's in the string. However, if the string comprises of only 0's it keeps one 0. [5]

OR

- Q6) a)** Write short notes on: [4]
- i) Reducibility
  - ii) Multi-tape Turing Machine

- b) Construct a Turing Machine for  $R=aba^*b$  [6]
- c) Design a TM that multiplies two unary numbers over  $\Sigma=\{1\}$ . Write simulation for the string  $11*111$ . [8]

- Q7) a)** Justify “Halting problem of Turing machine is undecidable” [8]
- b) Define and compare class P and class NP problem with suitable diagram [8]

OR

- Q8) a)** Explain in brief the term “recursively enumerable”. [6]
- b) Explain examples of problems in NP. [6]
- c) Differentiate between P Class and NP class. [4]



[6003]-347

**T.E. (Computer Engineering)**  
**THEORY OF COMPUTATION**  
**(2019 Pattern) (Semester-I) (310242)**

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate marks.
- 4) Assume suitable data, if necessary.

**Q1) a)** Give a Context Free Grammar for the following language. [9]

- i)  $L1 = \{a^i b^j c^k \mid i = j + k\}$  such that  $i, j, k > 0$
- ii)  $L2 = \{a^i b^j c^k \mid j = i + k\}$  such that  $i, j, k > 0$

b) Reduce the following grammar to Greibach Normal form. [9]  
 $S \rightarrow SS, S \rightarrow 0S1 \ 01$

OR

**Q2) a)** Show that the following grammar is ambiguous. [6]

 $S \rightarrow iCtS$ 
 $S \rightarrow iCtSeS$ 
 $S \rightarrow a$ 
 $C \rightarrow b$ 

b) Convert the following grammar to Chomsky Normal Form (CNF). [6]

 $G = (\{S\}, \{a, b\}, P, S)$ 
 $P = \{S \rightarrow aSa \mid bSb \mid a \mid b \mid aa \mid bb\}$ 

c) Consider the following grammar. [6]

 $E \rightarrow E + E \mid E - E \mid id$ 
Derive the string  $id-id*id$  using

- i) Leftmost derivation
- ii) Rightmost derivation

**Q3) a)** Find the transition rules of PDA for accepting a language  $L = \{w \mid w \in \{a, b\}^* \mid w \text{ is of the form } a^n b^n \text{ with } n \geq 1\}$  through both empty stack and final state and demonstrates the stack operation for the string  $aaabbb$ . [9]

P.T.O.

- b) Design a push down automation to recognize the language generated by the following grammar :  
 $S \rightarrow S + S \mid S \square S \mid 4 \mid 2$   
 Show the acceptance of the input string  $2+2*4$  by this PDA. [8]
- OR
- Q4)** a) What is NPDA? Construct a NPDA for the set of all strings over  $\{a,b\}$  with odd length palindrome. [9]  
 b) Design a push down automation to recognize the language generated by the following. [8]  
 $S \rightarrow S + S \mid S \square S \mid 4 \mid 2$   
 Show the acceptance of the input string  $2+2*4$  by this PDA.
- Q5)** a) Design a Turing Machine for the following language by considering transition table and diagram. [9]  
 i) TM that erases all non blank symbols on the tape where the sequence of non blank symbols does not contain any blank symbol B in between.  
 ii) TM that find 2's complement of a binary machine.  
 b) What is TM? Design TM to check well formedness of parenthesis. Expand the transition for  $(())()$  [9]
- OR
- Q6)** a) How turing machine can be use to compute the functions? Design turing machine for multiplication of two numbers. [9]  
 b) Elaborate the following terms. [9]  
 i) Universal Turing Machine (UTM)  
 ii) Recursively Enumerable Languages  
 iii) Halting problem of Turing Machine
- Q7)** a) Define and Compare Class P and Class NP Problem with suitable diagram. [9]  
 b) What do you mean by polynomial time reduction? Explain with suitable example. [8]
- OR
- Q8)** a) Explain Satisfiability Problem and SAT Problem and comment on NP Completeness of the SAT Problem. [9]  
 b) What makes a problem NP-Complete? How do we prove a problem is NP-complete? Are all decision problems NP-complete? [8]



Total No. of Questions : 8]

SEAT No. :

PB3778

[6262]-36

[Total No. of Pages :4

**T.E. (Computer Engineering)**  
**THEORY OF COMPUTATION**  
**(2019 Pattern) (Semester-I) (310242)**

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicates full marks.
- 4) Assume suitable data, if necessary.

**Q1) a)** What is context free Grammar? Define CFG. What are the capabilities of CFG? [8]

Give a context Free Grammar for the following language

$$L = \{ w \in \{a,b\}^* \mid w \text{ is a palindrome of odd length} \}.$$

- b) i) What is Derivation in CFG?
- ii) What is relation of parse tree for derivation in CFG?
- iii) What is leftmost derivation and Rightmost derivation?
- iv) Explain leftmost derivation and Rightmost derivation with parse tree. Derive the string a-b+c using leftmost derivation and Rightmost derivation for the CFG having production rule.

$$G = \{ S = S + S$$

$$S = S - S$$

$$S = a \mid b \mid c$$

}

[10]

OR

P.T.O.



**Q2) a)** When do we say that CFG is in Greibach Normal Form (GNF)? Explain the steps to convert CFG to GNF for following Grammars [12]

$$G1 = \{ S \rightarrow aAB \mid aB, A \rightarrow aA \mid a, B \rightarrow bB \mid b \}$$

$$G2 = \{ S \rightarrow aAB \mid aB, A \rightarrow aA \mid \epsilon, B \rightarrow bB \mid \epsilon \}$$

$$G3 = \{ S \rightarrow XB \mid AA$$

$$A \rightarrow a \mid SA$$

$$B \rightarrow b$$

$$X \rightarrow a \}$$

- b) i) What is ambiguity in CFG? What is relation of parse tree for finding ambiguity in CFG.
- ii) What is leftmost derivation and Rightmost derivation?
- iii) Explain leftmost derivation and Rightmost derivation and ambiguity for the CFG having production rule.

$$G = \{ S = aSb \mid SS$$

$$S = \epsilon \}$$

[6]

**Q3) a)** What is pushdown automata? Define PDA pictorially and mathematically with respect to input tape, stack, finite control and Instantaneous description.

Design a PDA for accepting a language  $\{a^n b^{2n} \mid n \geq 1\}$

[8]

- b) Construct a context free grammar which accepts  $N(A)$ , where [10]

$A = (\{q_0, q_1\}, \{0, 1\}, \{Z_0, Z\}, \delta, q_0, Z_0, \phi$  where  $\delta$  is given by

$$\delta(q_0, 1, Z_0) = \{(q_0, Z Z_0)\}$$

$$\delta(q_0, \epsilon, Z_0) = \{(q_0, \epsilon)\}$$

$$\delta(q_0, 1, Z) = \{(q_0, Z Z)\}$$

$$\delta(q_0, 0, Z) = \{(q_1, Z)\}$$

$$\delta(q_1, 1, Z) = \{(q_1, \epsilon)\}$$

$$\delta(q_1, 0, Z_0) = \{(q_0, Z_0)\}$$

OR

**Q4) a)** Design a PDA for accepting a language  $\{0^n 1^m 0^n \mid m, n \geq 1\}$ . [6]

b) Draw a PDA for the CFG given below: [6]

$$S \rightarrow aSb$$

$$S \rightarrow a \mid b \mid \epsilon$$

And simulate PDA to recognize "aaabb".

c) Design a push down automation to recognize the language generated by the following [6]

grammar:

$$S \rightarrow S + S \mid S * S \mid 4 \mid 2$$

Show the acceptance of the input string  $2 + 2 * 4$  by this PDA.

**Q5) a)** Elaborate the following terms with proper examples [4]

i) Universal Turing Machine (UTM)

ii) Recursively Enumerable Languages

b) Design a TM that multiplies two unary numbers over  $\Sigma = \{1\}$ . Write simulation for the string  $11 * 111$ . [7]

c) Construct a TM for the language  $L = \{0^n 1^n 2^n\}$  where  $n \geq 1$ . [6]

OR

**Q6) a)** Construct a TM for subtraction of two unary numbers  $f(a-b) = c$  where  $a$  is always greater than  $b$ . [5]

b) What is undecidability? How do we prove universal language is undecidable? What is the relation between undecidability and reducibility theory. [12]

**Q7) a)** What do you mean by polynomial time reduction? Explain with an example of SAT. [7]

b) Explain the following terms with respect to computations complexity with example. [10]

i) Solvable Vs Unsolvable problem

ii) Decidable Vs. Undecidable problem

iii) P Vs NP problem

OR

**Q8) a)** Explain in brief the term “recursively enumerable”. [6]

b) Explain examples of problems in NP. [6]

c) Differentiate between P class and NP class. [5]

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Total No. of Questions : 8]

SEAT No. :

P-7858

[Total No. of Pages : 3

[6180]-46A

T.E. (Computer Engineering)

THEORY OF COMPUTATION

(2019 Pattern) (Semester - I) (310242)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates :

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume suitable data, if necessary.

**Q1) a)** Check whether the string 10010 is a member of the language generated by following grammar by using Cocke-Younger-Kasami Algorithm- [9]

$S \rightarrow AB|BC$

$A \rightarrow BA|0$

$B \rightarrow CC|1$

$C \rightarrow AB|0$

**b)** Obtain grammar to generate the following language : [8]

$L = \{w : n_a(w) \bmod 2 = 0 \text{ where } w \in \{a, b\}^*\}$

i.e. Language of a and b in which number of number of a's in the string is either zero or in multiple of 2 only.

OR

**Q2) a)**

$S \rightarrow aB|bA$

$A \rightarrow a|aS|bAA$

$B \rightarrow b|bS|aBB$

Derive using Leftmost Derivation and Rightmost Derivation:

i) bbaaba      ii) aaabbb.

Draw parse tree for the same.

P.T.O.

b) Find context Free Grammar generating each of these languages. [8]

- i)  $L_1 = \{a^i b^j c^k \text{ such that } i = j+k \text{ where } i, j, k \geq 1\}$
- ii)  $L_2 = \{a^i b^j c^k \text{ such that } j = i+k \text{ where } i, j, k \geq 1\}$

**Q3) a)** Construct a PDA equivalent to following CFG [10]

i)

$X \rightarrow 0$

$X \rightarrow 0X$

$X \rightarrow 1XX$

$X \rightarrow XX1$

$X \rightarrow X1X$

ii)  $S \rightarrow BD|BC$

$D \rightarrow SC$

$C \rightarrow AA$

$B \rightarrow 0$

$A \rightarrow 1$

b) Design a PDA for a language  $L = \{a^n b^{2n} | n \geq 1\}$  [8]

OR

**Q4) a)** Construct a PDA accepting the language  $L = \{a^n b^m a^n | n, m \geq 0\}$  by null store. [6]

b) Design a PDA for a language  $L = \{XcX^r | X \in \{a,b\}^*$  and string  $X^r$  is the reverse of string  $X\}$ . [6]

c) Obtain a PDA to accept the language -

$L = \{w | w \in \Sigma^*, \Sigma = \{a,b\} \text{ and } n_a(w) = n_b(w)\}$  by final state [6]

**Q5) a)** Design a Turing machine for well formed parenthesis. [6]

b) Design a TM that accepts all strings over  $\{1,0\}$  with even number of 0's and even number of 1's. [8]

c) Construct TM that recognizes language over alphabet 0,1 such that string ends in 10. [4]

OR

- Q6)** a) Construct a TM to accept the language over  $\{0,1\}$  containing the substring 001. [6]  
 b) Design a TM to multiply a unary number by 2. [8]  
 c) Design Turing Machine for 1's Complement. [4]

- Q7)** a) What is post correspondence problem? Explain PCP with following instance of the set of the strings A and B. [8]

|    | A     | B   |
|----|-------|-----|
| 1. | 1     | 111 |
| 2. | 10111 | 10  |
| 3. | 10    | 0   |

- b) State and explain with suitable example [9]  
 i) Decidable Problem  
 ii) Undecidable Problem  
 iii) Church-Turing Thesis.

OR

- Q8)** a) What is reducibility in Computability Theory ? Explain in detail, the polynomial - time reduction approach for proving that a problem is NP-Complete. [8]  
 b) Explain with suitable example and diagrams [9]  
 i) Halting problem of TM  
 ii) Multitape TM  
 iii) Universal TM



Total No. of Questions : 8]

SEAT No. :

PC-1719

[Total No. of Pages : 3

[6353]-36

T.E. (Computer Engg.)

THEORY OF COMPUTATION

(2019 Pattern) (Semester - I) (310242)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume suitable data, if necessary.

**Q1) a)** Give Context Free Grammars for the following languages [9]

- i)  $L = \{ w \in \{a,b\}^* \mid w \text{ is string of starting with 'a' and ending with 'b'} \}$
- ii)  $RE = 0(0+1)^*01(0+1)^*1$
- iii)  $RE = (011+1)^*(01)^*$

**b)** Simplify the following grammar as [8]

- i) Eliminate Useless production

$S \rightarrow abS \mid abA \mid abB$

$A \rightarrow cd$

$B \rightarrow aB$

$C \rightarrow dc$

- ii) Eliminate Unit Production

$S \rightarrow Aa \mid B$

$A \rightarrow b \mid B$

$B \rightarrow A \mid a$

- iii) Eliminate the  $\epsilon$  Production

$S \rightarrow XYX$

$X \rightarrow 0X \mid \epsilon$

$Y \rightarrow 1Y \mid \epsilon$

OR

P.T.O.

**Q2) a)  $S \rightarrow aB \mid bA$**

**$A \rightarrow a \mid aS \mid bAA$**

**$B \rightarrow b \mid bS \mid aBB$**

Derive using Leftmost Derivation and Rightmost Derivation : i) bbaaba  
ii) aaabbb. Draw parse tree for the same. **[9]**

**b) Find context Free Grammar generating each of these languages. [8]**

i)  $L1 = \{ a^i b^j c^k \text{ such that } i = j + k \text{ where } i, j, k \geq 1 \}$

ii)  $L2 = \{ a^i b^j c^k \text{ such that } j = i + k \text{ where } i, j, k \geq 1 \}$

**Q3) a) i) Construct PDA for the given CFG, and test whether  $010^4$  is acceptable by this PDA.**

**$S \rightarrow 0BB$**

**$B \rightarrow 0S \mid 1S \mid 0$**

ii) Construct PDA for the given CFG, and test whether 'aaabb' is acceptable by this PDA.

**$S \rightarrow aSb$**

**$S \rightarrow a \mid b \mid \epsilon$**

**[10]**

**b) What is NPDA? Construct a NPDA for the set of all strings over  $\{a, b\}$  with odd length palindrome [8]**

**OR**

**Q4) a) Construct a PDA accepting the language  $L = \{a^n b^m a^n \mid n, m \geq 0\}$  by null store. [6]**

**b) Design a PDA for a language  $L = \{XcX^r \mid X \in \{a, b\}^*$  and string  $X^r$  is the reverse of string  $X\}$  [6]**

**c) Obtain a PDA to accept the language - [6]**

**$L = \{w \mid w \in \Sigma^*, \Sigma = \{a, b\} \text{ and } n_a(w) = n_b(w)\}$  by final state**

**Q5) a) Design the Turing for the function  $f(n) = 2n$  is computable. [9]**

**b) What are the different ways for extension of TM? Explain. Design TM for language  $L = \{a^m b^n \mid m < n\}$  [9]**



OR

**Q6) a)** Construct a TM to accept the language over  $\{0,1\}$  containing the substring 001. [6]

b) Design a TM to multiply a unary number by 2. [8]

c) Design Turing Machine for 1's Complement. [4]

**Q7) a)** What Traveling salesman problem? How to prove that Traveling salesman problem is NP Complete? [5]

b) What is post correspondence problem? Why is post correspondence problem undecidable? Explain PCP with following instance of the set of the strings A and B [12]

|    | A     | B   |
|----|-------|-----|
| 1. | 1     | 111 |
| 2. | 10111 | 10  |
| 3. | 10    | 0   |

OR

**Q8) a)** What is reducibility in Computability Theory ? Explain in detail, the polynomial time reduction approach for proving that a problem is NP Complete. [8]

b) State and explain with suitable example : [9]

i) Decidable Problem

ii) Undecidable Problem

iii) Church-Turing Thesis.



Total No. of Questions : 8]

SEAT No. :

PD4242

[Total No. of Pages : 2

[6403]-36

**T.E. (Computer Engineering)**  
**THEORY OF COMPUTATION**  
**(2019 Pattern) (Semester - V) (310242)**

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume suitable data if necessary.

**Q1) a)** Give Context Free Grammars for the following languages and show the derivation for given string. **[12]**

- i)  $L = \{ w \in \{a,b\}^* \mid w \text{ is string of starting with 'a' and ending with 'b' } \}$   
show the derivation for "ababab"
- ii)  $L = a^n b^{2n}$  where  $n \geq 1$ . Show the derivation for "aabbbb"
- iii)  $RE = (0+1)^*$  Show the derivation for "0110"

**b)** Reduce the following grammar to Greibach Normal form. **[6]**

$S \rightarrow AA \mid 0$   
 $A \rightarrow SS \mid 1$

OR

**Q2) a)** Convert the following grammar to CNF. **[8]**

$S \rightarrow aSa \mid bSb \mid A \mid \epsilon$   
 $A \rightarrow a \mid b \mid \epsilon$

**b)** In the grammar, convert the given production rule into GNF form. If any production rule in the grammar is not in GNF form, convert it. **[10]**

$S \rightarrow XB \mid AA$   
 $A \rightarrow a \mid SA$   
 $B \rightarrow b$   
 $X \rightarrow a$

**Q3) a)** i) Construct PDA for the given CFG, and test whether  $010^4$  is acceptable by this PDA. **[10]**

$S \rightarrow 0BB$   
 $B \rightarrow 0S \mid 1S \mid 0$

ii) Construct PDA for the given CFG, and test whether 'aaabb' is acceptable by this PDA.

$S \rightarrow aSb$   
 $S \rightarrow a \mid b \mid \epsilon$

**b)** Construct PDA to accept  $L = \{ a^n b^n \mid n \geq 1 \}$  through empty stack. **[4]**

**c)** Convert the given PDA to CFG  $S \rightarrow 0S1 \mid 00 \mid 11$  **[4]**

OR

P.T.O.

- Q4)** a) Construct Pushdown automata for  $L = \{0^n 1^m 2^{(n+m)} \mid m, n \geq 0\}$  [6]  
 b) NPDA for accepting the language  $L = \{a^{2m} b^{3m} \mid m \geq 1\}$  [6]  
 c) NPDA for accepting the language  $L = \{a^i b^j c^k d^1 \mid i=j \text{ or } j=1, i \geq 1, j \geq 1\}$  [6]

- Q5)** a) Write short notes on with suitable diagrams. [12]  
 i) Reducibility  
 ii) Multi-tape Turing Machine  
 iii) Multi-head Turing Machine  
 iv) Two-way infinite Tape Turing Machine:  
 v) Multi-tape Multi-head Turing Machine  
 b) Construct a TM for subtraction of two unary numbers  $f(a-b) = c$  where  $a$  is always greater than  $b$ . Explain the logic of building this Turing machine. [5]

OR

- Q6)** a) Draw a Turing Machine to increment a binary number by 1 and demonstrate with any example. [6]  
 b) Obtain a Turing Machine to accept the language containing strings of  $a$ 's and  $b$ 's that do not end with  $abb$ . [7]  
 c) Construct a TM for the language  $L = \{0^n 1^n 2^n\}$  where  $n \geq 1$  [4]

- Q7)** a) What Minimum spanning tree problem? Prove that finding MST by using Kruskal's algorithm is in class P. [5]  
 b) What is post correspondence problem? Why is post correspondence problem undecidable? Explain PCP with following instance of the set of the strings A and B. [12]

|    | A     | B   |
|----|-------|-----|
| 1. | 1     | 111 |
| 2. | 10111 | 10  |
| 3. | 10    | 0   |

OR

- Q8)** a) Define and Compare Class P and Class NP Problem with suitable diagram. [8]  
 b) State and explain with suitable example. [9]  
 i) Decidable Problem  
 ii) Undecidable Problem  
 iii) Church-Turing Thesis

